

SpeakMate AI - Detailed Technical Documentation

React

React is a JavaScript library used to build interactive user interfaces.

Why Used:

- Reusable components
- Easier maintenance
- Dynamic UI updates

Role in SpeakMate AI:

- Dashboard
- Speech Recorder
- Progress Charts
- Interview Panels

Without React, the frontend would become difficult to maintain and scale.

Next.js

Next.js is a full-stack framework built on top of React.

Why Used:

- Provides frontend and backend in one framework
- Built-in routing
- API routes
- Performance optimization

Role in SpeakMate AI:

- Frontend pages
- Backend APIs
- Authentication integration
- Routing

Without Next.js, a separate Express backend would be required.

PostgreSQL

PostgreSQL is a relational database used to store structured data.

Why Used:

- Strong support for relationships
- Reliable and scalable
- ACID compliant

Role in SpeakMate AI:

- User information
- Assessments
- Progress history
- Daily challenges

Chosen over MongoDB because the project contains highly relational data.

Prisma

Prisma is an ORM (Object Relational Mapper).

Why Used:

- Simplifies database operations
- Type safety
- Better developer experience

Role in SpeakMate AI:

- Database communication layer
- Query execution
- Data management

Prisma acts as a translator between JavaScript and PostgreSQL.

NextAuth

Authentication framework for Next.js.

Why Used:

- Secure authentication
- OAuth integration
- Session management

Role:

- Google Login

- GitHub Login
- Protected routes
- User sessions

Groq

Groq is an AI inference platform.

Why Used:

- Extremely fast model inference
- Lower latency

Role:

- Hosts Whisper Large V3 Turbo
- Hosts Llama 3.3 70B

Provides near real-time AI responses.

Whisper Large V3 Turbo

Speech recognition model.

Purpose:

Convert audio recordings into text.

Role:

Audio -> Transcript

Why Used:

- High accuracy
- Multi-language support
- Fast inference through Groq

Llama 3.3 70B Versatile

Large Language Model with 70 billion parameters.

Role:

- Communication analysis
- Root cause detection
- Coaching tips
- Motivational feedback
- Diagnosis generation

Why Used:

- Strong reasoning capability
- Fast on Groq
- Cost effective

Acts as the communication coach of SpeakMate AI.

Python + Librosa

Librosa is an audio analysis library.

Why Used:

Text alone cannot accurately measure confidence.

Role:

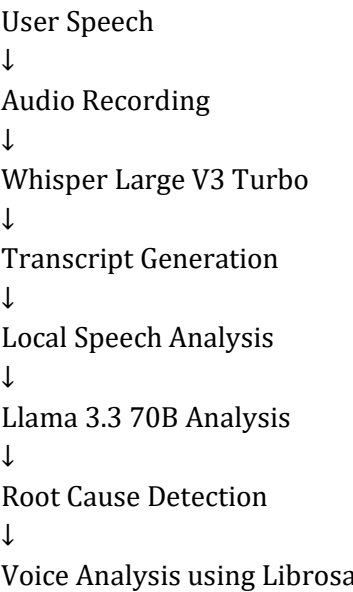
- Pitch analysis
- Energy analysis
- Pause detection
- Speaking rate detection
- Nervousness estimation
- Confidence estimation

Provides real voice intelligence.

AI Models Usage Table

Component	Technology
Speech Recording	MediaRecorder API
Speech Transcription	Whisper Large V3 Turbo
Transcript Analysis	Llama 3.3 70B Versatile
Root Cause Detection	Llama 3.3 70B Versatile
Coaching Tips	Llama 3.3 70B Versatile
Motivational Feedback	Llama 3.3 70B Versatile
Voice Analysis	Python + Librosa
Confidence Detection	Llama + Librosa
Nervousness Detection	Librosa
Pause Detection	Librosa
Speaking Rate Detection	Librosa
Score Aggregation	JavaScript Logic
Data Storage	PostgreSQL
ORM	Prisma
Authentication	NextAuth
Backend	Next.js API Routes
Frontend	React + Next.js

Complete AI Workflow



↓

Score Merging

↓

Feedback Generation

↓

PostgreSQL Storage

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Dashboard Visualization